Functional Skills
Mathematics Level 1

Practice

OFFICIAL MFS MODEL SOLUTION

60 / 60 Marks (100% Pass)

- Assessment: Practice Assessment 1 (FSMI07P)
- Complete step-by-step worked models
- Official Mark Scheme Allocations [M1 / A1]
- Exam Board: Open Awards (Ofqual Regulated)

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LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION A - QUESTION AND ANSWER PAPER
NON-CALCULATOR – 30 MINUTES
PRACTICE ASSESSMENT 1 (FSM107P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** includes Task 1. You **must not** use a calculator for this section.

  Total marks available: 15. Time limit: 30 minutes
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section

  Total marks available: 45. Time limit: 1 hour and 30 minutes

For Section A you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor

INTERNET ACCESS IS NOT PERMITTED AND YOU MUST NOT USE A CALCULATOR

The invigilator will stop the assessment after 30 minutes. You must hand in this question and answer paper at this point.

The invigilator will then hand out **Section B** and a non-scientific calculator. You will then have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section A** at the end. Once you have handed in this question and answer paper, you will not be able to check this again.
7. If you use extra paper, write your name, learner number and the question number you are answering on it and securely attach it to this question and answer paper.

Learner name:	
Learner number:	
Centre number:	
Signature:	
Today's date:	

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Section A

Question 1

Which calculation would result in the same answer as 20 x 0.5? (1 mark)

Multiplying by 0.5 is the same as finding half (÷ 2):

20 × 0.5 = 10 --> 20 ÷ 2 = 10

40 ÷ 2 | 40 x 2 | 20 x 2 | 20 ÷ 2 [1 mark]

Write your answer in this box:

20 ÷ 2 ✓

Question 2

Step 1: Convert pence to pounds by dividing by 100 (£1 = 100p):

What is 1340 pence written in pounds and pence? (1 mark)

Show your calculations and/or workings out here:

1340 ÷ 100 = 13.40

= £13.40 [1 mark]

Write your answer in this box:

£13.40 ✓

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Question 3

Put the following numbers in order of size: (1 mark)

Compare place values starting from highest digit (hundred thousands):

232,149 < 322,419 < 332,491 232149 322419 332491

[1 mark]

Write your answer in this box:

232149, 322419, 322491, 332491

Question 4

Calculate 256.4 ÷ 1000 (1 mark)

Step 1: Dividing by 1000 moves decimal point 3 places to the left:

Show your calculations and/or workings out here:

256.4 ÷ 1000 = 0.2564

[1 mark]

0.2564

Write your answer in this box:

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Question 5

Which **two** of the following are equivalent to 20%? (2 marks)

$\frac{2}{5}$ 0.02 $\frac{1}{5}$ $\frac{4}{5}$ 0.8 0.2

Step 1: Convert 20% to a simplified fraction:

Show your calculations and/or workings out here:

20% = 20 / 100 = 1 / 5 [1 mark]

Step 2: Convert 20% to a decimal:

20% = 20 ÷ 100 = 0.2 [1 mark]

Write your answers in the box:

1/5 and 0.2

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Question 6

Omar wants to buy a tablecloth and some napkins.

Tablecloth	£14.99
Napkins	£6.99

He pays with £30.00.

How much change will he receive? Give your answer to the nearest pound.

(2 marks)

Show your calculations and/or workings out here:

Step 1: Calculate total cost of items:

£14.99 + £6.99 = £21.98

Step 2: Subtract total from £30.00 to find exact change:

£30.00 - £21.98 = £8.02 [M1]

Step 3: Round change to the nearest whole pound (£):

£8.02 rounds down to £8.00 [A1]

Write your answer in this box:

£8

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Question 7

Omar is making cheesecakes. He has this recipe:

1 Cheesecake (serves 4)

240g digestive biscuits
100g butter
600g soft cheese
100g icing sugar
284ml double cream

Step 1: Find how many cheesecakes are needed for 16 servings:

16 servings ÷ 4 servings per recipe = 4 batches [M1]

Omar has 750g of digestive biscuits and enough of all the other ingredients. He thinks he has enough digestive biscuits to make 16 servings.

Step 2: Calculate total biscuits required for 4 batches:

Is Omar correct? Give a reason for your answer. (4 marks)

240g × 4 = 960g biscuits required [M1 A1]

Show your calculations and/or workings out here:

Step 3: Compare with biscuits available:

Omar only has 750g. Short by: 960g − 750g = 210g

No, Omar is not correct. ✓

He needs 960g of biscuits for 16 servings, but only has 750g (he is 210g short). [A1]

Write your answer and reason in this box:

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Question 8

The table below shows the number of visitors to a museum each day in one week.

Mon	Tue	Wed	Thur	Fri	Sat	Sun
32		46	48	55	72	61

The mean number of people per day is 50.
How many people visited on Tuesday? (3 marks)

Step 1: Calculate total visitors needed across 7 days for mean of 50:

Show your calculations and/or workings out here:

7 days × 50 visitors = 350 visitors in total [M1]

Step 2: Add up visitors on the known 6 days:

32 + 46 + 48 + 55 + 72 + 61 = 314 visitors [M1]

Step 3: Subtract from total to find Tuesday visitors:

Tuesday = 350 - 314 = 36 visitors [A1]

36 visitors ✓

Write your answer in this box:

[End of Section A]

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LEVEL 1 FUNCTIONAL SKILLS QUALIFICATION IN MATHEMATICS

SECTION B - QUESTION AND ANSWER PAPER CALCULATOR – 1 HOUR 30 MINUTES PRACTICE ASSESSMENT 1 (FSM107P)

Do not open this paper until you are told to do so by the invigilator.

Overall assessment marks available: **60**
Overall assessment time limit: **2 HOURS**

There are **TWO** Sections to this assessment:

- **Section A** – please ensure you have handed in Section A before beginning Section B
- **Section B** includes Task 2, 3 and 4. You can use a non-scientific calculator for this section.

Total marks available: 45. Time limit: 1 hour and 30 minutes.

For Section B you need:

- This question and answer paper
- A pen with black or blue ink
- A pencil
- A ruler and a protractor
- A non-scientific calculator

INTERNET ACCESS IS NOT PERMITTED

You now have a further 1 hour and 30 minutes to complete **Section B**.

Instructions

1. Please sign and date below to confirm that your details are correct and that you have understood the instructions.
2. Read each task and question carefully.
3. Remember to show all your workings out clearly.
4. The number of marks available for each question is shown in brackets. Use these marks to guide you on how long to spend on each question.
5. Answer **all** questions using the space provided on this question and answer paper.
6. If you have time, check your work for **Section B** at the end.
7. If you use extra paper, write your name, learner number and the question number you are answering on it, and securely attach it to this question and answer paper.
8. At the end of this section (**Section B**), hand in this question and answer paper and all notes to the invigilator.

Learner name:	
Learner number:	
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Signature:	
Today's date:	

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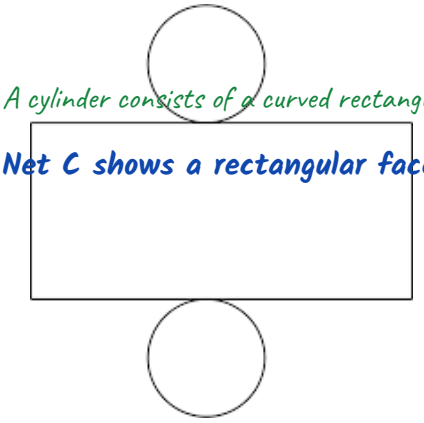
Section B

Question 9

Which of the following nets makes a cylinder? (1 mark)

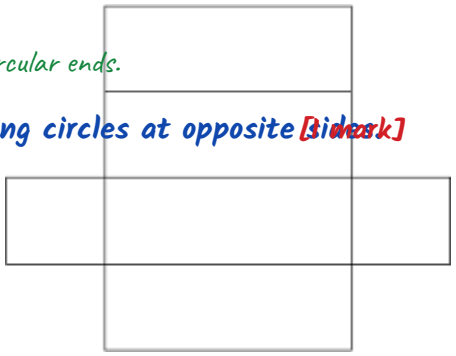
A



A cylinder consists of a curved rectangular surface and two circular ends.

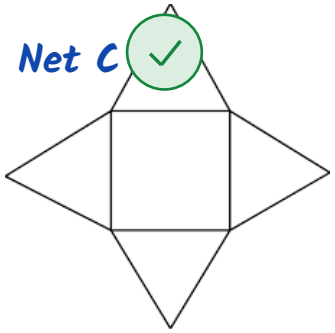
Net C shows a rectangular face with two matching circles at opposite sides. [1 mark]

B



C

Net C ✓



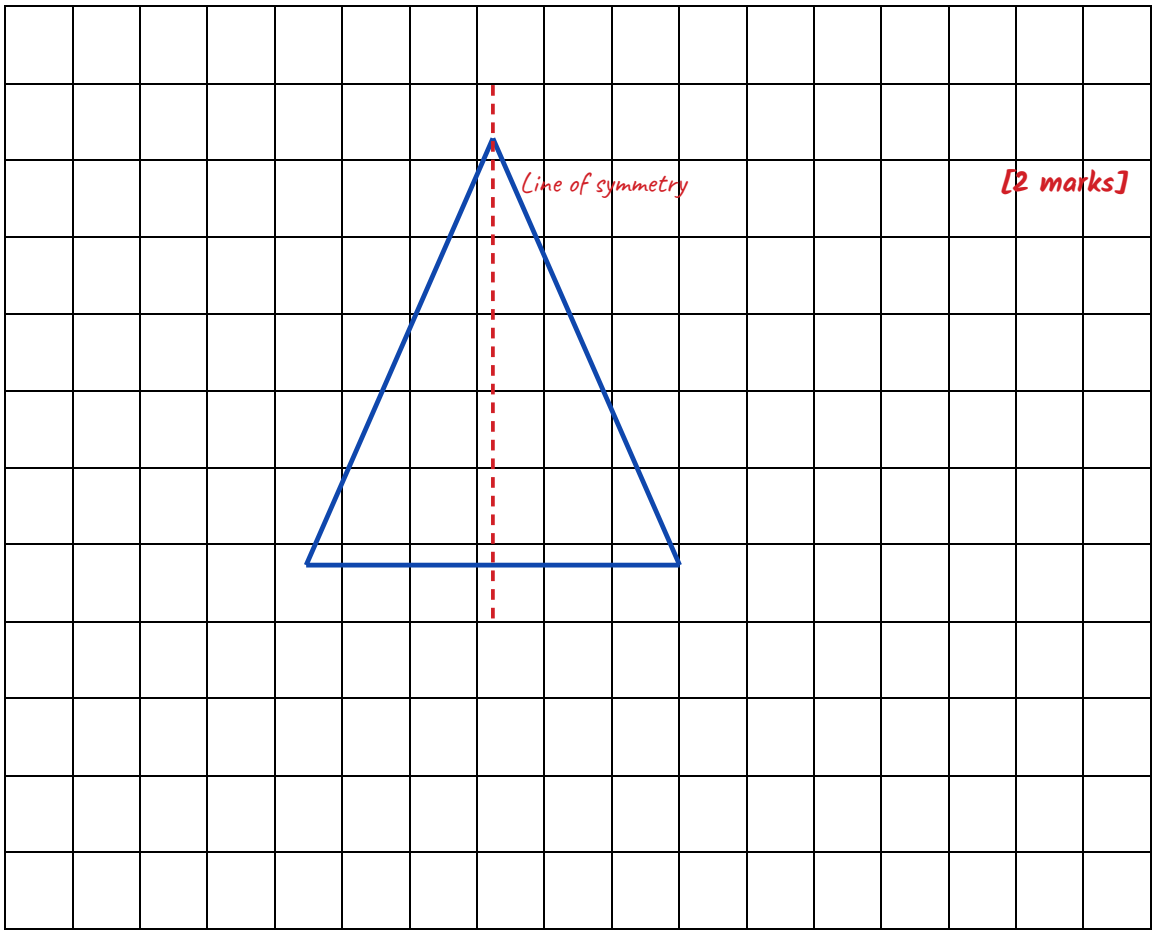
D

Write your answer in this box:

Question 10

Draw a triangle with at least one line of symmetry. Show the line of symmetry on your triangle. (2 marks)

Isosceles triangle drawn with vertical line of symmetry:



[2 marks]

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Question 11

Anna works at an animal rescue centre. There were 6160 animals rescued last year.

- ¼ of these were cats
- 20% of these were dogs
- The rest were other animals.

How many other animals were rescued last year? (3 marks)

Step 1: Calculate number of rabbits (1/4 of 6160):

Show your calculations and/or workings out here:

6160 ÷ 4 = 1540 rabbits [M1]

Step 2: Calculate number of cats (20% of 6160):

6160 × 0.20 = 1232 cats [M1]

Step 3: Subtract rabbits and cats from total to find dogs:

Total rabbits + cats = 1540 + 1232 = 2772

Dogs = 6160 - 2772 = 3388 dogs [A1]

3388 dogs ✓

Write your answer in this box:

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Question 12

Last year the animal rescue centre rescued 2520 dogs and cats in the ratio of 3:2.

This year the number of dogs rescued has increased by 15%.

How many dogs have been rescued this year? (5 marks)

Show your calculations and/or workings out here:

Step 1: Divide 2520 animals in the ratio 3:2 (dogs : cats):

~~Total ratio parts = 3 + 2 = 5 parts~~

1 part = 2520 ÷ 5 = 504 animals [M1]

Step 2: Find number of dogs rescued last year (3 parts):

Dogs = 504 × 3 = 1512 dogs [M1 A1]

Step 3: Calculate 15% expected increase for this year:

Increase = 1512 × 0.15 = 226.8

Total dogs = 1512 + 226.8 = 1738.8 --> 1739 dogs [M1 A1]

Write your answer in this box:

1739 dogs ✓

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Question 13

Anna needs to put some rabbits in the rabbit run for exercise.

Each rabbit needs 2.5m^2 of space in the rabbit run.

The diagram below shows a plan of the rabbit run.

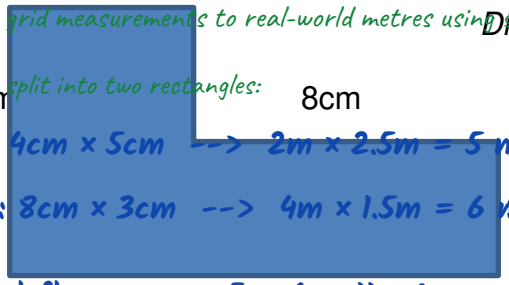

Diagram not drawn accurately

Step 1: Convert grid measurements to real-world metres using Scale (2cm = 1m)

Compound shape split into two rectangles:

Rectangle 1: 4cm × 5cm --> 2m × 2.5m = 5 m²

Rectangle 2: 8cm × 3cm --> 4m × 1.5m = 6 m² [M1]

Step 2: Total floor area = 5 + 6 = 11 m² [A1]

Scale: 2cm on the plan represents 1m on the rabbit run.

Step 3: Calculate maximum rabbits accommodated (each rabbit needs 2.5 m²):

What is the maximum number of rabbits that can be put into the rabbit run?

11 m² ÷ 2.5 m² = 4.4 rabbits --> 4 whole rabbits [M1 A1]

(4 marks)

Show your calculations and/or workings out here:

4 rabbits ✓

Write your answer in this box:

For Markers Use Only

Question 14

Put the following fractions in order of size: (1 mark)

$\frac{1}{2}$ $\frac{1}{4}$ $\frac{3}{4}$ $\frac{7}{12}$

Show your calculations and/or workings out here:

Step 1: Convert all fractions to common denominator 12:

1/4 = 3/12, 1/2 = 6/12, 7/12, 3/4 = 9/12 [1 mark]

1/4, 1/2, 7/12, 3/4

Write your answer in this box:

Question 15

Calculate 23^2 (1 mark)

Show your calculations and/or workings out here:

Step 1: Square the number (multiply by itself):

$23^2 = 23 \times 23 = 529$ [1 mark]

Write your answer in this box:

529

Question 16

There are two blue sweets and seven red sweets in a bag.

What is the probability of picking out a blue sweet at random? Express your answer as a fraction.

(1 mark)

Show your calculations and/or workings out here:

Step 1: Total number of sweets = 2 blue + 7 red = 9 sweets

Probability of picking a blue sweet = 2 / 9 [1 mark]

Write your answer in this box:

2 / 9 ✓

Question 17

Leah works as a nurse on a children's ward.

The table below shows the weight of each child on the ward.

Children	Weight
Child A	16.06kg
Child B	16.5kg
Child C	16.51kg
Child D	16.451kg

Step 1: Child C has the greatest weight: 16.51 kg [MI]

Step 2: Convert kg to grams (× 1000):

16.51 × 1000 = 16,510 g [MI]

Leah calculates the daily amount of medicine to give a child using this formula:

Amount of medicine in millilitres = Weight of child in grams ÷ 454 x 0.75

Step 3: Apply medicine formula: Weight in grams ÷ 454 × 0.75

16510 ÷ 454 × 0.75 = 27.2742... ml [MI]

How much medicine will she give to the **heaviest** child on the ward? Give your answer to one decimal place.

Step 4: Round to 1 decimal place (1 d.p.):

(4 marks)

27.2742... --> 27.3 ml [AI]

Show your calculations and/or workings out here:

27.3 ml ✓

Write your answer in this box:

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Question 18

The following table shows the number of visitors to the children's ward each day for one week.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
23	19	26	19	23	15	8

Anna thinks the range of visitors is 18. Is she correct? (2 marks)

Step 1: Calculate the range (Highest value - Lowest value).

Show your calculations and/or workings out here:

Highest = 26, Lowest = 8

Range = 26 - 8 = 18 visitors [M1]

Step 2: Compare with target range (target is less than 20):

18 < 20, so the range is indeed less than 20. [A1]

Write your answer in this box:

Yes ✓

The range is 18 (26 - 8), which is less than 20.

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Question 19

The table below shows the time in minutes that 30 patients waited to be seen by a doctor.

5	10	32	45	20	24	17	11	39	9
15	28	36	40	22	7	28	39	27	10
4	45	38	27	35	26	13	8	22	11

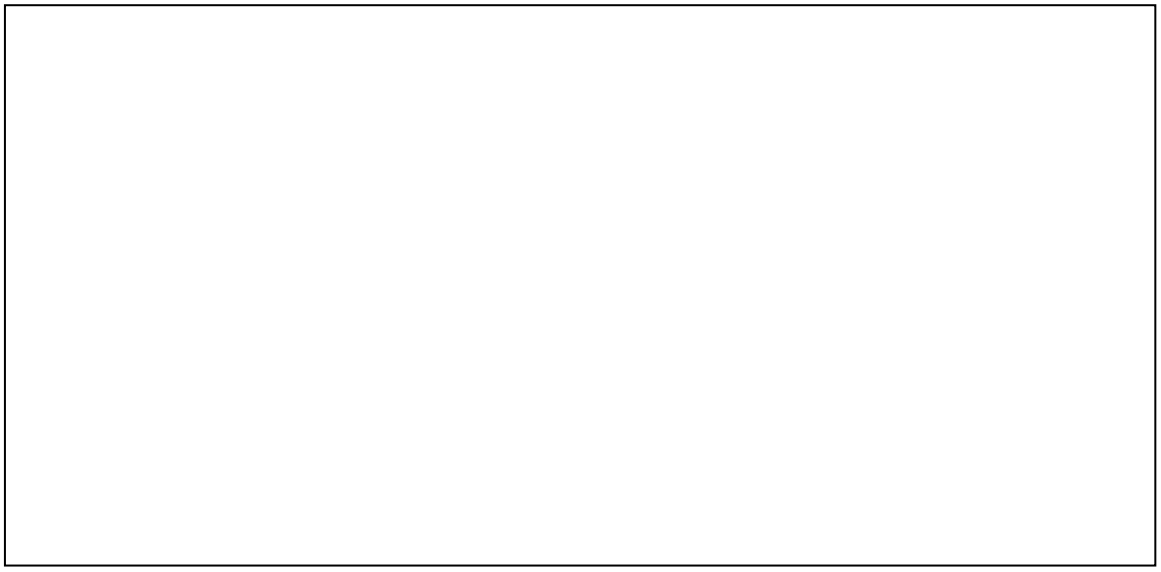
Step 1: Group waiting times into equal, non-overlapping intervals: [MI]

Create a suitable frequency table below to show the data. (3 marks)

Waiting Time (minutes)	Frequency
0 - 10	7
11 - 20	5
21 - 30	8
31 - 40	7
41 - 50	3

[MI AI]

Show your calculations and/or workings out here:



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Question 20

Callum is a porter at the hospital. He puts the following 3 items into a box ready to carry them to another room.

Item 1	5.25kg
Item 2	4.02kg
Item 3	3.08kg

The box cannot contain items weighing more than 15kg in total.

Callum wants to add another item weighing 2.68 kg.

Can he put the extra item in the box without it weighing over 15kg?
Give a reason for your answer. (3 marks)

Show your calculations and/or workings out here:

Step 1: Calculate total weight of the 4 items when added together:

5.25 + 4.02 + 3.08 + 2.68 = 15.03 kg [M1 A1]

Step 2: Compare total weight to the 15 kg safe lifting limit:

15.03 kg > 15 kg (it exceeds the limit by 0.03 kg / 30g).

No, the box is over the weight limit. [A1]

Total weight is 15.03 kg, which is 0.03 kg over the 15 kg limit.

Write your answer and reason in this box:

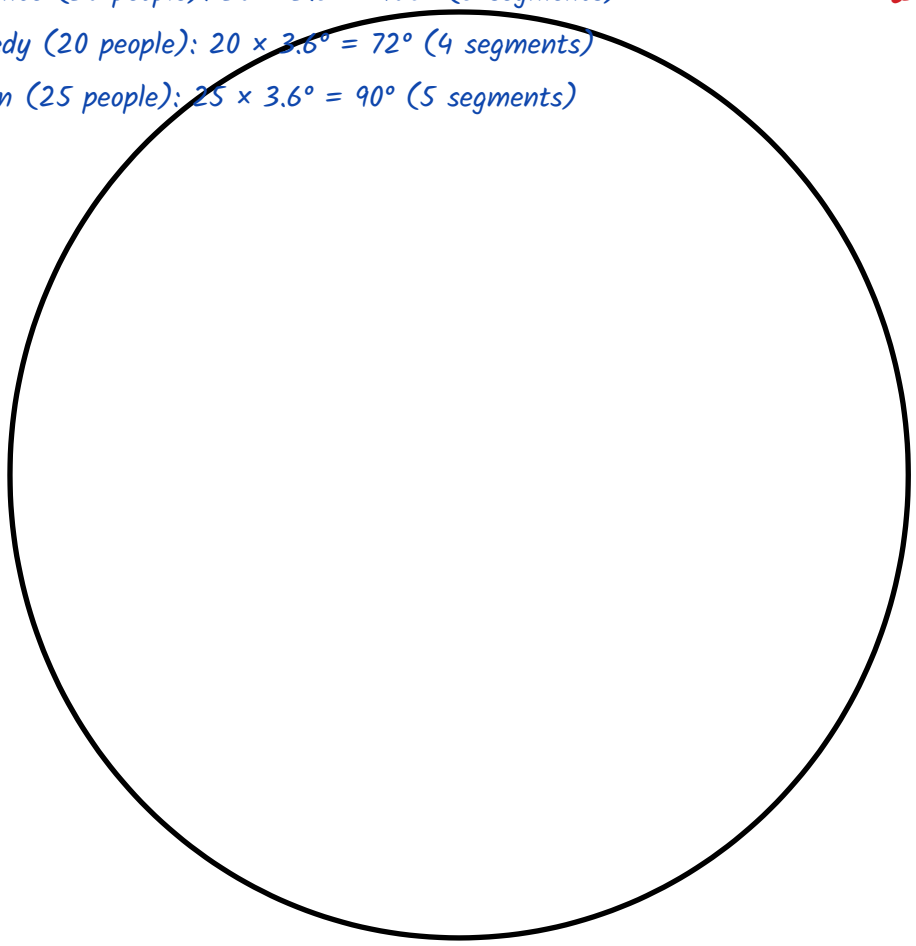
Question 21

Favourite Types of Film for 100 People					
Type of Film	Sci Fi	Drama	Romance	Comedy	Action
Number of people	20	5	30	20	25

Step 1: 100 people total. Each person = 360° ÷ 100 = 3.6° (or 1 segment = 18° for 20 segments):

Draw a pie chart to represent this data. (3 marks)

- Sci Fi (20 people): 20 × 3.6° = 72° (4 segments)
- Drama (5 people): 5 × 3.6° = 18° (1 segment)
- Romance (30 people): 30 × 3.6° = 108° (6 segments) **[3 marks]**
- Comedy (20 people): 20 × 3.6° = 72° (4 segments)
- Action (25 people): 25 × 3.6° = 90° (5 segments)



Question 22

Ali works as a maintenance man. He wants to put fence panels around a children's play area.

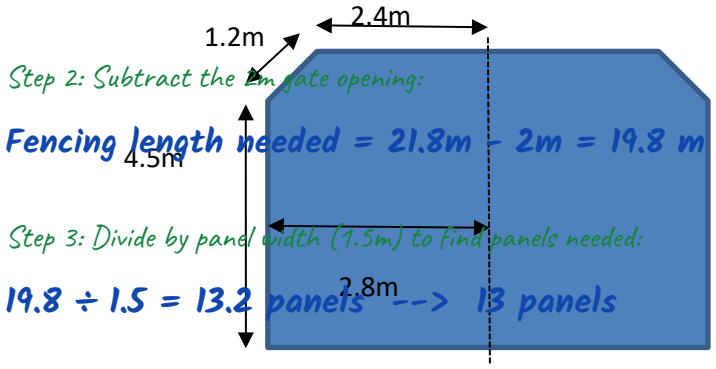
The fence panels are 1.5 metres wide. He will need to leave a gap of at least 200cm for a gate.

The outdoor space has one line of symmetry shown by a dotted line on the diagram below.

Step 1: Calculate total perimeter of the play area:

Perimeter = (2.4 + 1.2 + 4.5 + 2.8) × 2 = 21.8 m

Diagram not to sale [M1]



Step 2: Subtract the 2m gate opening:

Fencing length needed = 21.8m − 2m = 19.8 m [A1]

Step 3: Divide by panel width (1.5m) to find panels needed:

19.8 ÷ 1.5 = 13.2 panels --> 13 panels [M1 A1]

How many whole fence panels does Ali need to buy? (4 marks)

Show your calculations and/or workings out here:

13 panels ✓

Write your answer in this box:

Question 23

Samantha works for a delivery company. She needs to make a delivery of 255 boxes.

Her van has a width of 2m.

The boxes that she needs to transport are cubes with lengths of 40cm.

Is Samantha's van large enough to hold the 255 boxes? Give a reason for your answer.

(4 marks)

Step 1: Convert all dimensions to consistent units (metres):

Van: Length = 4000mm = 4m, Width = 2000mm = 2m, Height = 2m [M1]

Box: 400mm = 0.4m each side

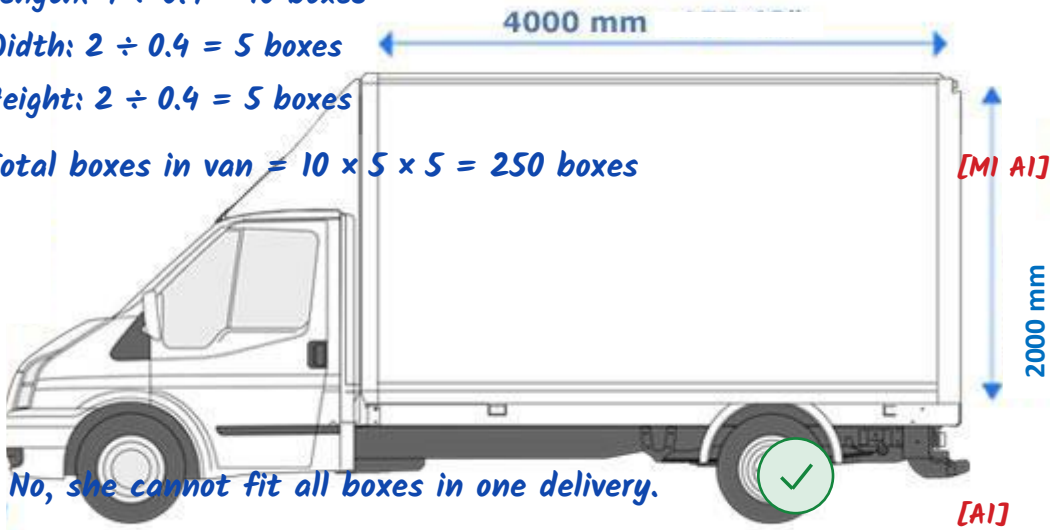
Step 2: Calculate how many boxes fit along each dimension:

Length: 4 ÷ 0.4 = 10 boxes

Width: 2 ÷ 0.4 = 5 boxes

Height: 2 ÷ 0.4 = 5 boxes

Total boxes in van = 10 × 5 × 5 = 250 boxes [M1 A1]



No, she cannot fit all boxes in one delivery. [A1]

She can only fit 250 boxes in the van, but has 255 boxes to deliver.

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Show your calculations and/or workings out here:

Write your answer and reason in this box:

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Question 24

Samantha has a delivery round each day. The time taken for each delivery in Samantha's round is shown below.

Delivery 1	75 minutes
Delivery 2	84 minutes
Delivery 3	23 minutes
Delivery 4	53 minutes

Step 1: Calculate total delivery time in minutes:

75 + 84 + 23 + 53 = 235 minutes [M1]

Samantha finds a different route which will reduce the time taken to make all her deliveries by $\frac{2}{5}$

Step 2: Reduce time by 2/5:

2/5 of 235 = (235 ÷ 5) x 2 = 47 x 2 = 94 minutes reduction

New time = 235 - 94 = 141 minutes (or 235 x 3/5 = 141) [M1 A1]

How long will it take Samantha to complete all her deliveries using the new route?

Give your answer in hours and minutes. (4 marks)

Show your calculations and/or workings out here:

Step 3: Convert 141 minutes into hours and minutes:

141 ÷ 60 = 2 hours remainder 21 minutes

= 2 hours and 21 minutes [A1]

2 hours and 21 minutes ✓

Write your answer in hours and minutes in this box:

[End of assessment]

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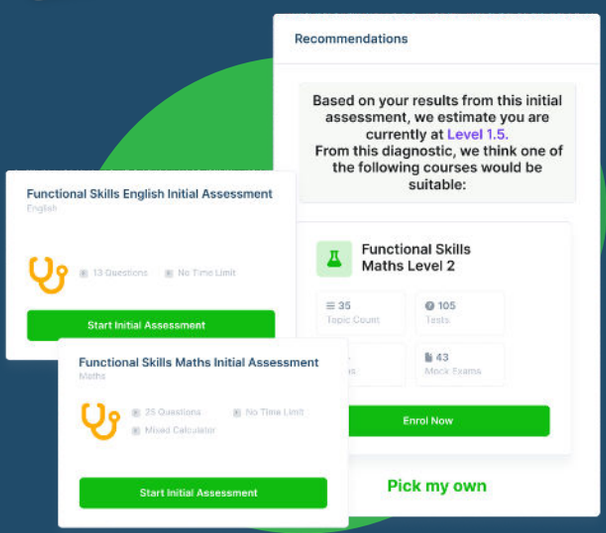

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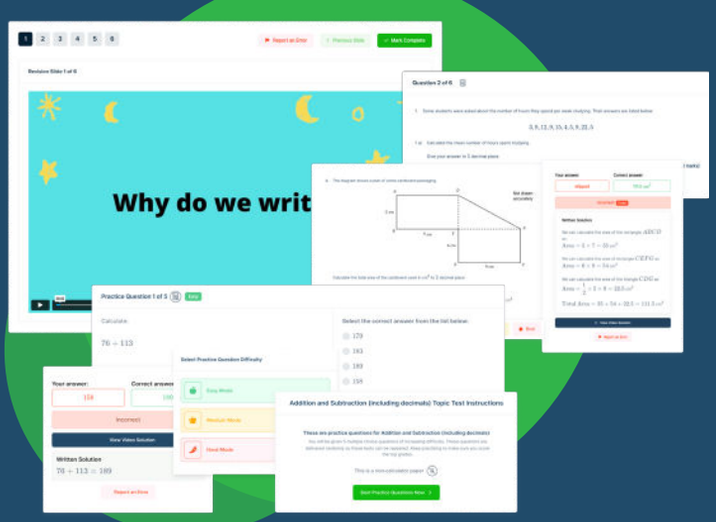

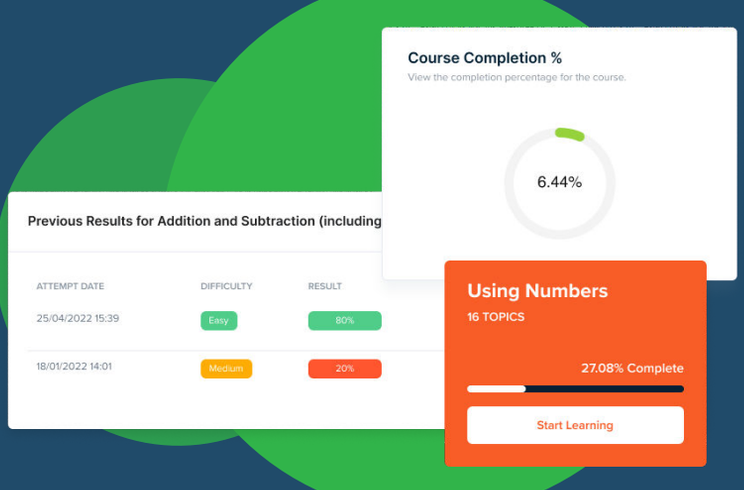

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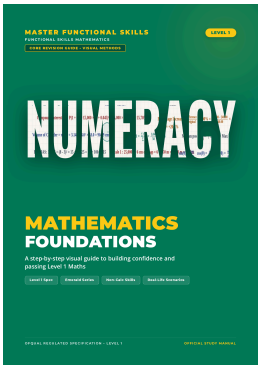

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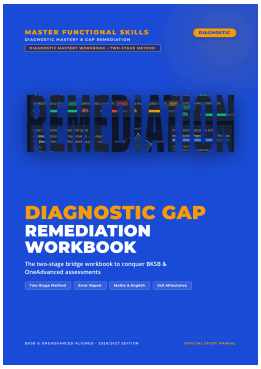

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